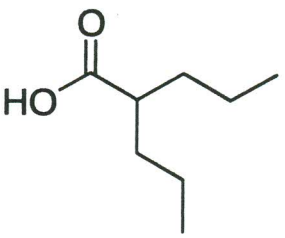
Cumulative Exam
Analytical Chemistry
February 13, 2015

The questions below have been used for in-class problem solving (Feb 6, 2015) and the first exam (February 13, 2015) for the CHEM 2263 course.

1. (25 points) You have synthesized valproic acid, a medication used to treat epileptic seizure, for the first time in the laboratory (see image to the right hand side). A requirement for FDA approval is to provide the acid's pK_a value in your documentation to the FDA for drug approval. You make a solution that is 0.1 M in its formal concentration and then measure the pH. The pH of this solution at equilibrium is 2.9. What is the pK_a for valproic acid?



2. (50 points) Recent work by Schoenfisch and colleagues reports the creation of new nitric oxide (NO) electrochemical sensors.[1] NO is a neuromodulator that is of significant biomedical interest. These sensors would be targeted for using as a means to make a direct measurement within a tissue, an in vivo chemical analysis. The following table lists values for log $K_{(NO, j)}$, the selectivity coefficient, where j is the interfering species.

Interference Tested	**log $K_{(NO, j)}$**
Ascorbic Acid	-5.74
Dopamine	-6

a) If NO levels in the brain are roughly 100 pM, what concentrations for each of the interferences would be needed to give a response from the electrode that would appear as if 100 pM NO was present.

b) If dopamine levels are roughly 10 nM in the synapse, could this NO sensor be used to reliably measure NO if NO levels are 100 pM? Explain why or why not.

c) Dopamine is stored in vesicles. Many researchers have found that dopamine concentrations in a vesicle are typically 5 mM. Could this NO electrode be used for measurement of dopamine near a neuronal vesicle that is releasing dopamine at 5 mM in the presence of 100 pM NO? Explain why or why not.

d) Ascorbic acid is typically 1 μM in the brain. Would ascorbic acid pose a problem for interference for NO detection (100 pM NO)? Explain why or why not.

[1] Further information for this work can be found at the following web site.
http://www.ncbi.nlm.nih.gov/pubmed/18714964/

3. (25 points) Consider removing Hg_2^{2+} from Cu^+.[2] Coal-fired power plants are the single largest emitter of mercury so we may want to know how much mercury is in coal. {This is a hypothetical example.} Let's say we dissolve up the coal and the concentrations of Hg_2^{2+} and Cu^+ are both 0.01 M. {Again, hypothetical as normally we have no idea the concentrations in an unknown sample.}

Could 99% of the Hg_2^{2+} be removed from the Cu^+ by adding excess Br^-?

K_{sp} (CuBr) $= 5 \times 10^{-9}$
K_{sp} (Hg_2Br_2) $= 5.6 \times 10^{-23}$

[2] Note that Hg_2^{2+} is a polyatomic ion. It doesn't dissolve into 2 Hg atoms.